

EBN122 CLASS TEST / KLASTOETS 2 (23 SEP 2016)**ANSWER SHEET / ANTWOORD BLAD**

Surname:		Student number:							
Initials:		AFR (Eng 3-6)	ENG (Eng 3-1)	ENG (Eng 3-6)					

Total:	/15
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1	$V_{0(30V)} =$	
2	$V_0 =$	
3	$V_{th} =$	$R_{th} =$
4	$R_N =$	
5	$V_0 =$	
6	$V_I =$	
7	$I_S =$	

DO NOT remove this page from the test.

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15 Marks/Punte

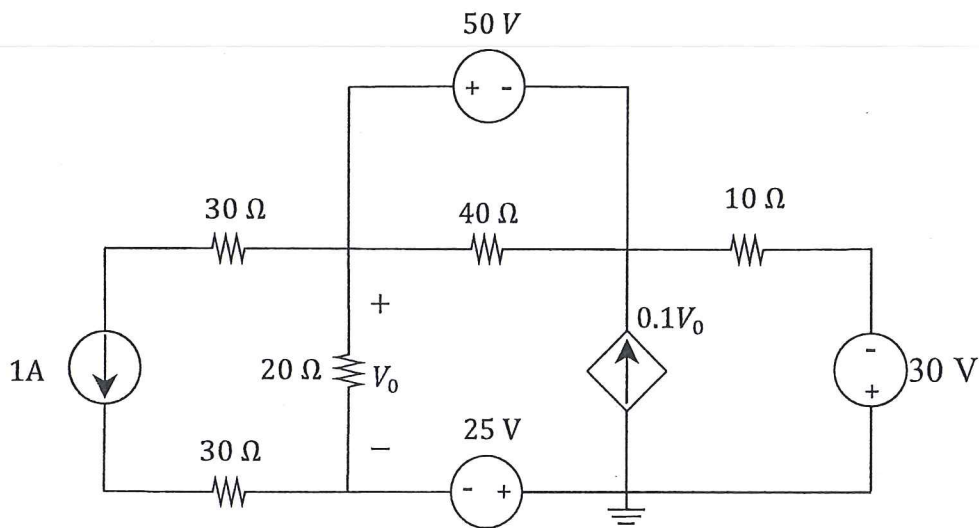
30 Minutes/Minute

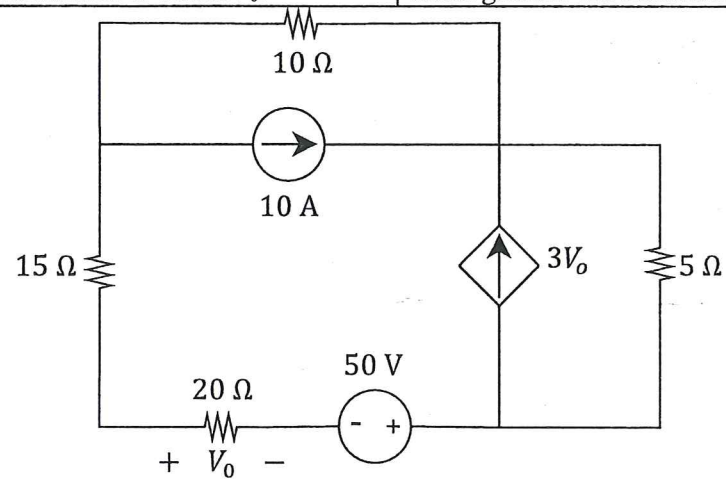
Question 1 [2]

Using superposition, find the individual contribution of the 30V voltage source to V_o in the circuit below.

Vraag 1 [2]

Gebruik superposisie, en bepaal die individuele bydra van die 30 V spanningsbron tot V_o .



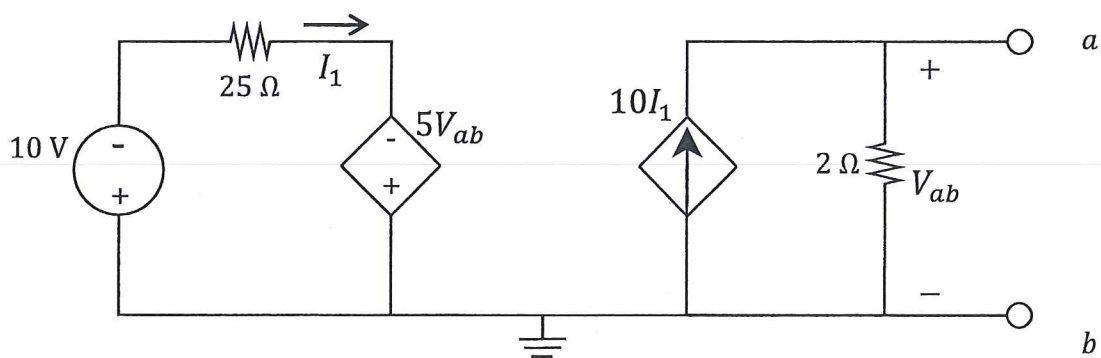
Question 2 [2]Make use of source transformation to obtain V_o .**Vraag 2 [2]**Maak gebruik van brontransformatie om V_o te bepaal.

Question 3 [4]

Find the Thevenin equivalent with respect to terminals a - b .

Vraag 3 [4]

Bepaal die Thevenin ekwivalent met verwysing tot terminale a - b .

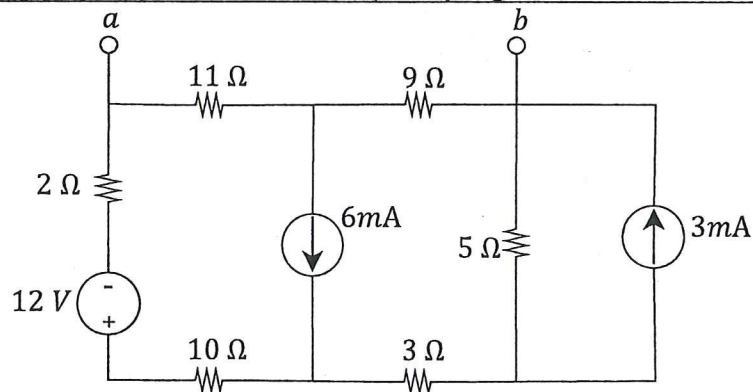


Question 4 [2]

Determine the Norton equivalent resistance R_N with respect to terminals $a-b$.

Vraag 4 [2]

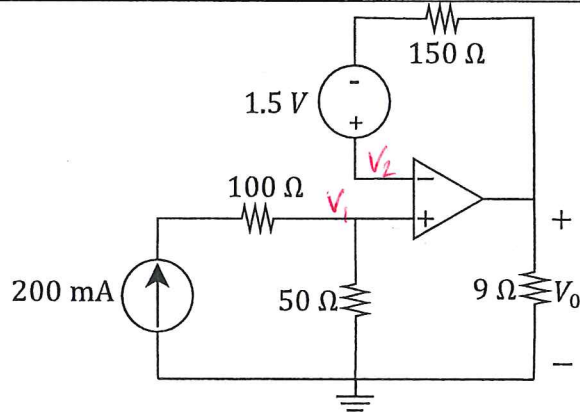
Bepaal die Norton ekwivalente weerstand R_N met verwysing tot terminale $a-b$.

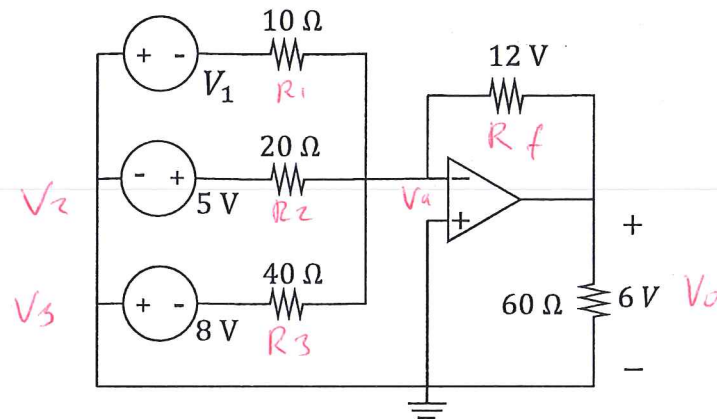
**Question 5 [2]**

Obtain V_o for the op amp circuit below.

Vraag 5 [2]

Bepaal V_o vir onderstaande kringbaan.



Question 6 [2]Obtain V_I for the op amp circuit below.**Vraag 6 [2]**Bepaal V_I vir onderstaande kringbaan.**Question 7 [2]**Obtain I_S for the op amp circuit below.**Vraag 7 [2]**Bepaal I_S vir onderstaande kringbaan.